

Chemical Kinetics

Subject / Theme :

Date :

☐ Sat ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri

→ Introduction

→ Order & Molecularity

→ Definition and examples $\Rightarrow$ Zero order reaction

$\Rightarrow$ 1st order reaction

$\Rightarrow$ 2nd order reaction

$\Rightarrow$ Pseudo unimolecular reaction.

→ Integrated rate equation for 1st order reaction

→ Integrated rate equation for 2nd order reaction.

→ Activation energy and Activated complex.

→ Collision Theory.

☐ Introduction:-

Chemical kinetics may also be called reaction kinetics or simply kinetics. The rate of a chemical reaction usually has units of sec^{-1} .

In 1864 Peter Waage and Cato Guldberg pioneered the development of chemical kinetics by formulating the law of mass action, which states the speed of a chemical reaction is proportional to the quantity of the reacting substances.

→ Reaction rate is the rate at which the reactants of a chemical reaction form the product.

→ Reaction rates are expressed as concentration per unit time.

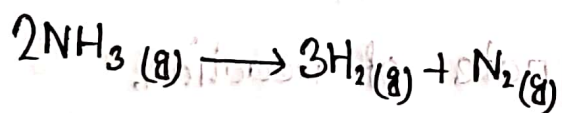
Order:-

Order of reaction is the total power of concentration of reactants which is proportional to the experimental facts.

Order of reaction is an experimental fact which depends on experiment, observation and conclusion.

→ Order of reaction never be fraction

Example:-



order → Zero order reaction.

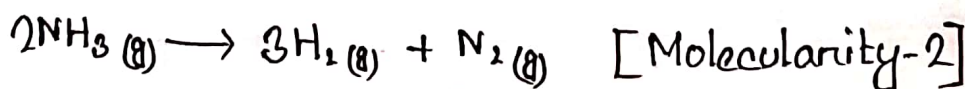
Molecularity:-

Molecularity of a reaction is the number of atoms, ions or molecules taking part in a reaction as in chemical equation.

→ Molecularity is a theoretical concept which depends on chemical equation partly.

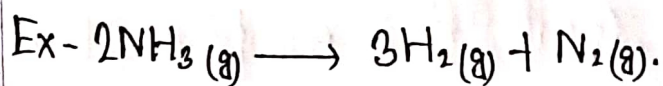
→ Molecularity can be fraction

Example:-



Zero Order Reaction:-

The reaction whose rate is independent of the concentration has no effect on the speed of the reaction is called zero order reaction.



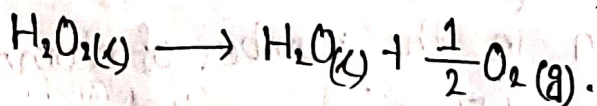
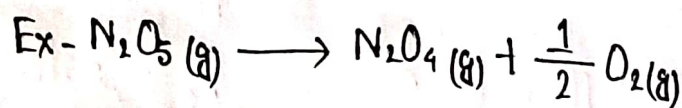
1st order reaction:-

If a reaction rate depends on a single reactant and the value of the exponent is one, then the reaction is said to be 1st order.

The reaction whose rate is proportional to first power of concentration of one reactant is called the 1st order reaction.

According to defⁿ, rate of reaction,

$$R = \frac{-dc}{dt} = k \times c^1 = kC$$



Second Order Reaction:-

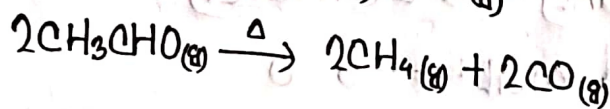
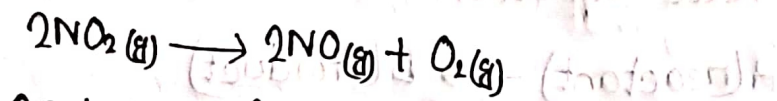
The reaction whose rate is proportional to second power of concentration of one reactant or two reactants is called second order reaction.

According to the defⁿ, rate of reaction, $r = \frac{-dc}{dt} = Kc^2$

→ A reaction is said to be second order when the overall order is two. The rate of a second order reaction may be proportional to one concentration squared or to the product of two concentrations.

→ Two types: i) $2(A) \text{ (reactant)} \longrightarrow \text{Products}$
ii) $A + B \longrightarrow \text{Products}$.

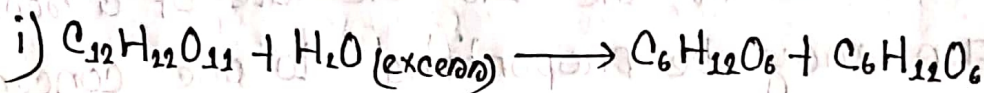
Example:-



Pseudo Unimolecular Reaction:-

It is a reaction which seems to be second degree but actually a first order reaction. One of the reactants is in excess which makes its importance negligible and to follow the first order rate is called Pseudo Unimolecular Reaction.

Example:-

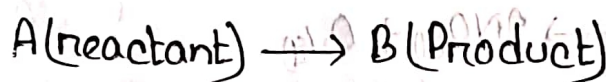


$$\text{Rate} = k [C_{12}H_{22}O_{11}]$$



$$\text{Rate} = k [CH_3COOC_2H_5]$$

☐ Integrated rate equation for 1st order reaction:-



Initially $t=0$, $a \text{ mol L}^{-1} \longrightarrow 0$

After $t=t$, $(a-x) \text{ mol L}^{-1} \longrightarrow x$

According to law of mass action, rate of 1st order reaction at that time will be:

$$\frac{dx}{dt} \propto (a-x)^1$$

$$\Rightarrow \frac{dx}{dt} = k(a-x) \quad [k \text{ is a constant}]$$

$$\Rightarrow \frac{dx}{(a-x)} = k dt$$

Integrating both sides,

$$\int \frac{dx}{a-x} = \int k dt$$

$$\Rightarrow -\ln(a-x) = kt + c \quad \text{--- (i)}$$

When,

$$t=0, x=0 \text{ from (i),}$$

$$-\ln(a-0) = k \times 0 + c$$

$$\Rightarrow -\ln a = c$$

$$\Rightarrow c = -\ln a$$

putting the value of c in eqn (i)

$$-\ln(a-x) = kt - \ln a$$

$$\Rightarrow kt = \ln a - \ln(a-x)$$

$$\Rightarrow kt = \ln \frac{a}{a-x}$$

$$\Rightarrow k = \frac{1}{t} \ln \frac{a}{a-x}$$

$$\Rightarrow k = \frac{2.303}{t} \log \frac{a}{a-x}$$

Unit

Here,

a = initial concentration of reactant A.

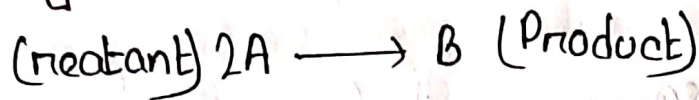
x = decrease concentration of reactant A, increased concentration of product B.

k = Rate constant.

$a-x$ = concentration of reactant at time t

t = Time of reaction at which reaction proceed.

☐ Integrated equation for 2nd order reaction:-



Initially $t=0$, $a \text{ mol l}^{-1} \longrightarrow 0$

After $t=t$, $(a-x) \text{ mol l}^{-1} \longrightarrow x$

According to law of mass action, at any time t the rate of reaction will be,

$$\frac{dx}{dt} \propto (a-x)^2$$

$$\Rightarrow \frac{dx}{dt} = k(a-x)^2 \quad [k \text{ is a constant}]$$

$$\Rightarrow \frac{dx}{(a-x)^2} = k dt$$

Integrating both side we get,

$$\int \frac{dx}{(a-x)^2} = \int k dt$$

$$\Rightarrow \frac{1}{a-x} = kt + c \quad \text{--- (i)}$$

When $t=0$, $x=0$ from (i)

$$\frac{1}{a-0} = k \times 0 + c$$

$$\Rightarrow c = \frac{1}{a}$$

putting the value of c in eqn ①

$$\frac{1}{a-x} = kt + \frac{1}{a}$$

$$\Rightarrow kt + \frac{1}{a} = \frac{1}{a-x}$$

$$\Rightarrow \frac{1}{a-x} - \frac{1}{a} = kt$$

$$\Rightarrow kt = \frac{a - a + x}{a(a-x)}$$

$$\Rightarrow kt = \frac{x}{a(a-x)}$$

$$\Rightarrow k = \frac{1}{t} \cdot \frac{x}{a(a-x)}$$

unit



Reaction rate

$$k = \frac{1}{t} \ln \frac{a}{a-x}$$

unit

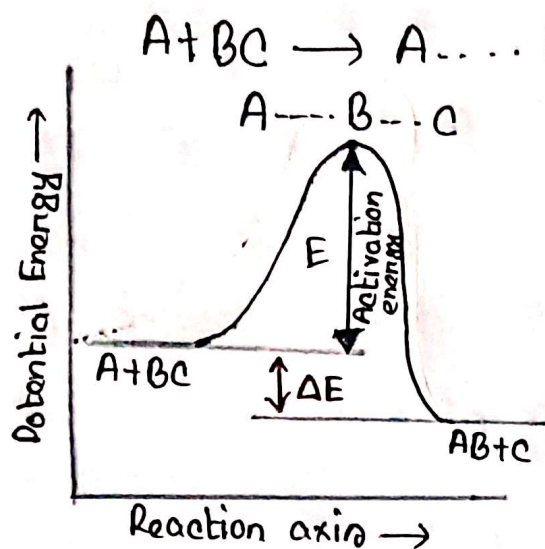
rate constant of reaction
(or) rate of reaction
(or) rate of reaction

fraction of molecules with activation energy

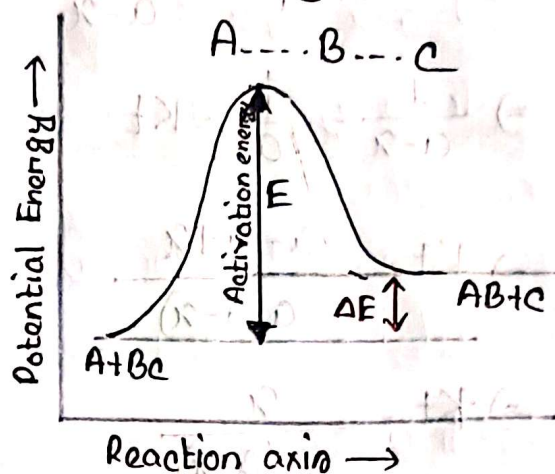
unit

Collision theory is used to explain the rate of reaction. It is based on the idea that molecules must collide with sufficient energy and proper orientation to undergo a chemical reaction. The rate of reaction is proportional to the number of effective collisions per unit time.

☐ Activation energy and activated complex:-



Exothermic Reaction



Endothermic Reaction

☐ Collision Theory:-

$$K = P Z e^{-\frac{E_a}{RT}}$$

Where,

K = rate constant of reaction

P = steric factor (condition no 3)

Z = collision rate (condition no 1)

$e^{-\frac{E_a}{RT}}$ = fraction of molecules with activation energy.

Collision theory was successfully applied to explain the bimolecular reactions in gaseous phase but later it was found to be applicable also to be the reaction in solution.

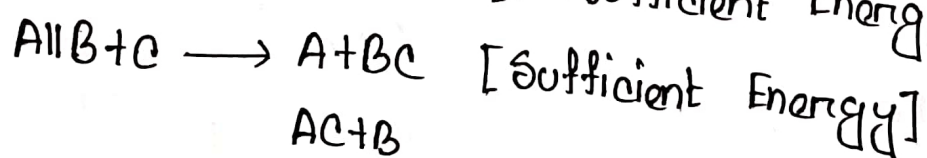
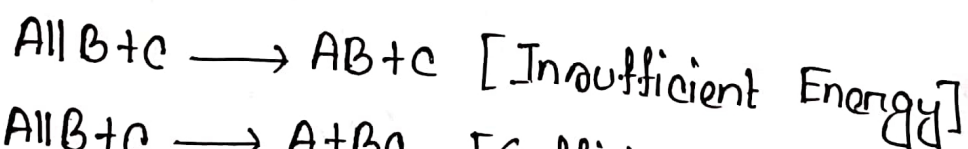
For reaction to take place,

The 1st condition $\rightarrow$ The particles (atom, ion or molecules) taking part in a reaction must collide with one another.

The 2nd condition $\rightarrow$ The collision must take place with a minimum amount of energy.

The 3rd condition $\rightarrow$ The collisions must take place with definite orientation. Arrhenius equation is in fact the mathematical expression of these conditions.

According to the 1st and 2nd conditions the reacting particles must collide with one another with minimum energy. Then some of the old chemical bonds will be broken down and ~~an~~ new bonds will be formed.



Example:-

